THE THREE CADENCES OF PHYSICAL AI

Multi-Rate Asynchronous Hierarchy: Untrusted Semantic Intent vs. Continuous Trajectory Chunks vs. Bare-Metal Reflex

SYSTEM 2 · 0.5–2 Hz

Semantic Intent Engine

Linux MPU / Neural Accelerator (NPU)

Workload & Representation:

Open-world cognitive deliberation

- Multimodal VLM (7B–70B parameters)
- 3D spatial affordance grounding
- Kinematic reachability screening
- Untrusted candidate proposals

Latency Tail Profile:

P50 ≈ 650 ms · P99 ≈ 1,450 ms

Subject to OS jitter and memory stalls.

OUTPUT CONTRACT: EXPIRING LEASE

`L_intent = (T_target, Δt_TTL, δ_drift)`

- Metric SE(3) target coordinate frame
- Time-to-Live lease (e.g., TTL = 800 ms)
- Continuous drift abort threshold

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SYSTEM 1.5 · 20–50 Hz

Action Chunk Planning

Linux MPU / Neural Accelerator (NPU)

Workload & Representation:

Receding-horizon trajectory synthesis

- Diffusion Policy / ACT decoders
- Predicts H=16 action waypoints
- C^2 jerk-continuous spline blend
- Exponential temporal ensembling

Latency Tail Profile:

P50 ≈ 22 ms · P99 ≈ 44 ms

Amortizes inference delay over horizon H.

OUTPUT CONTRACT: TRAJECTORY CHUNK

`A_(t:t+H) ∈ R^(H x D_act)`

- Lock-free shared SRAM ring buffer
- Bounded derivative jerk: $|j| \leq j_{max}$
- Asynchronous state re-anchoring on veto

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SYSTEM 1 · 1000 Hz MCU

Real-Time Safety Reflex

Bare-Metal / FreeRTOS Microcontroller

Workload & Representation:

Deterministic mathematical safety barrier

- Control Barrier Functions: $h(x) \geq 0$
- Quadratic program solve in < 65 µs
- Dynamic stopping audit: $d_{stop} \leq d_{clear}$
- Zero heap allocation (zero malloc)

Deterministic Execution Budget:

Loop = 175 µs · Slack = 825 µs

Hard real-time: P99 ≤ 1.000 ms.

OUTPUT CONTRACT: MOTOR GATE PWM

`PWM_register <- u_safe(t)`

- Sole physical authority over motor coils
- Fallback: IEC 60204-1 Cat 0/1/2 safe stop
- Hardware watchdog trips on MPU stall